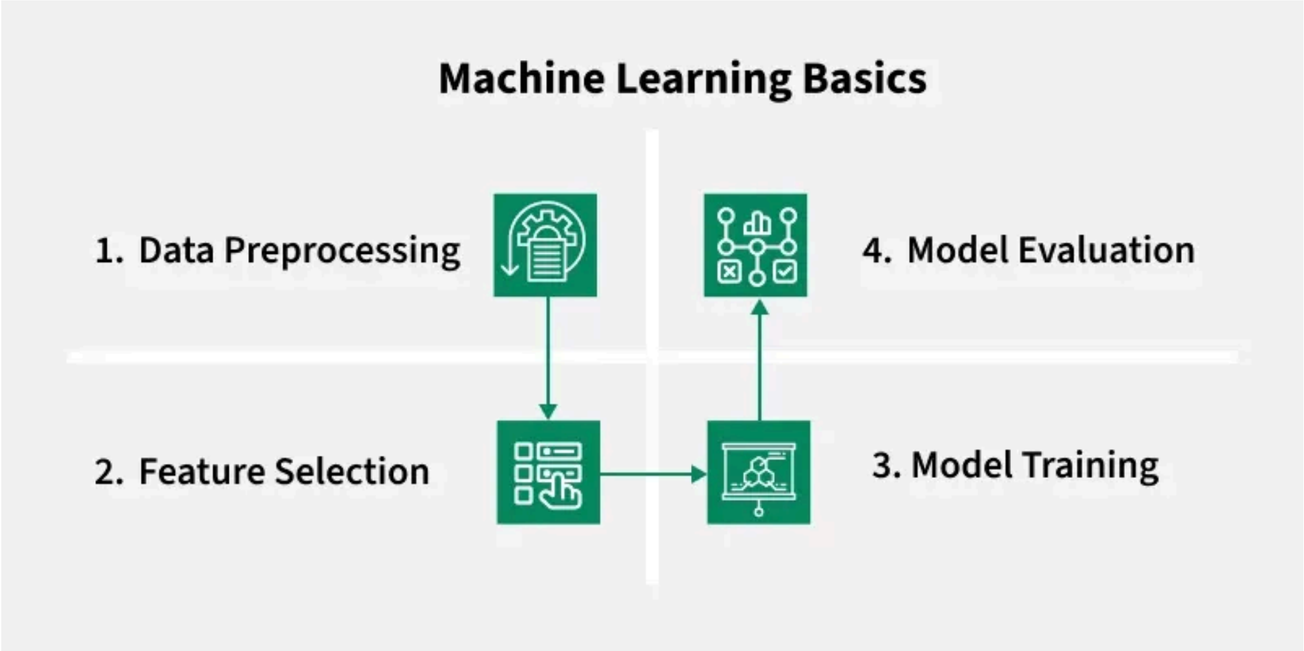
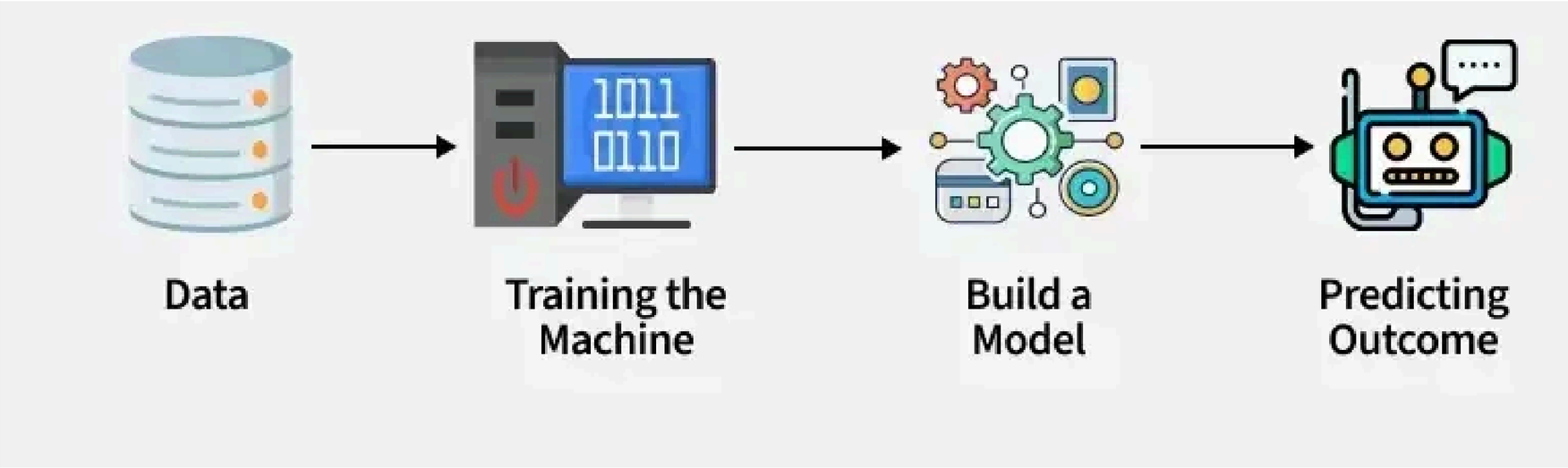
Machine Learning

Machine learning is a branch of Artificial Intelligence that focuses on developing models and algorithms that let computers learn from data without being explicitly programmed for every task. In simple words, ML teaches the systems to think and understand like humans by learning from the data.



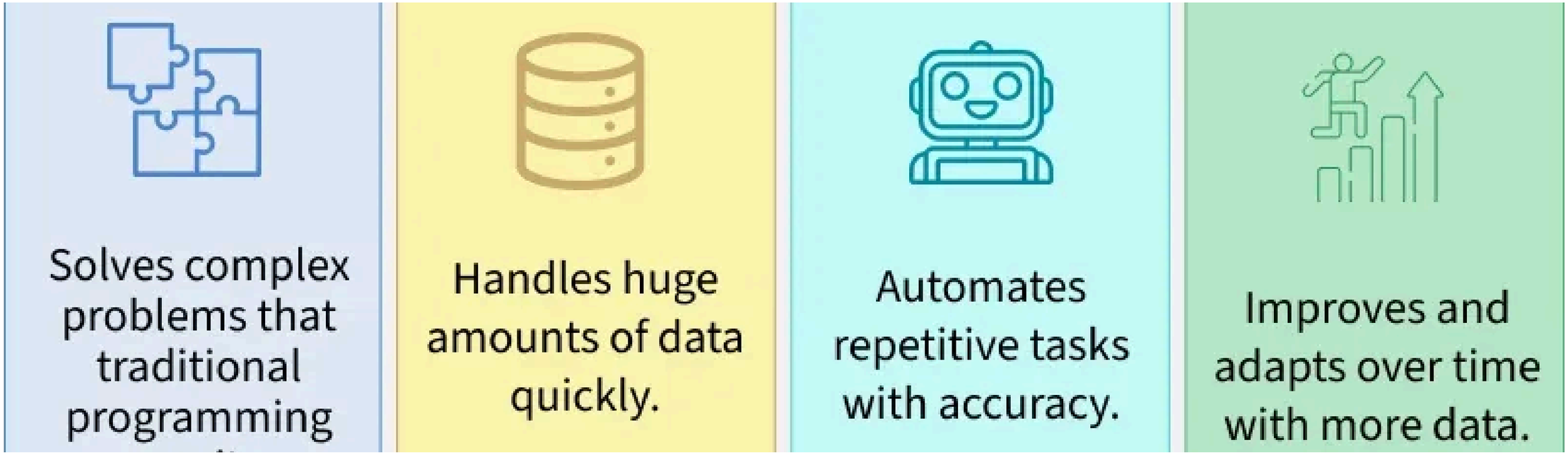
Introduction of machine learning

It involves feeding data into algorithms to identify patterns and make predictions on new data.

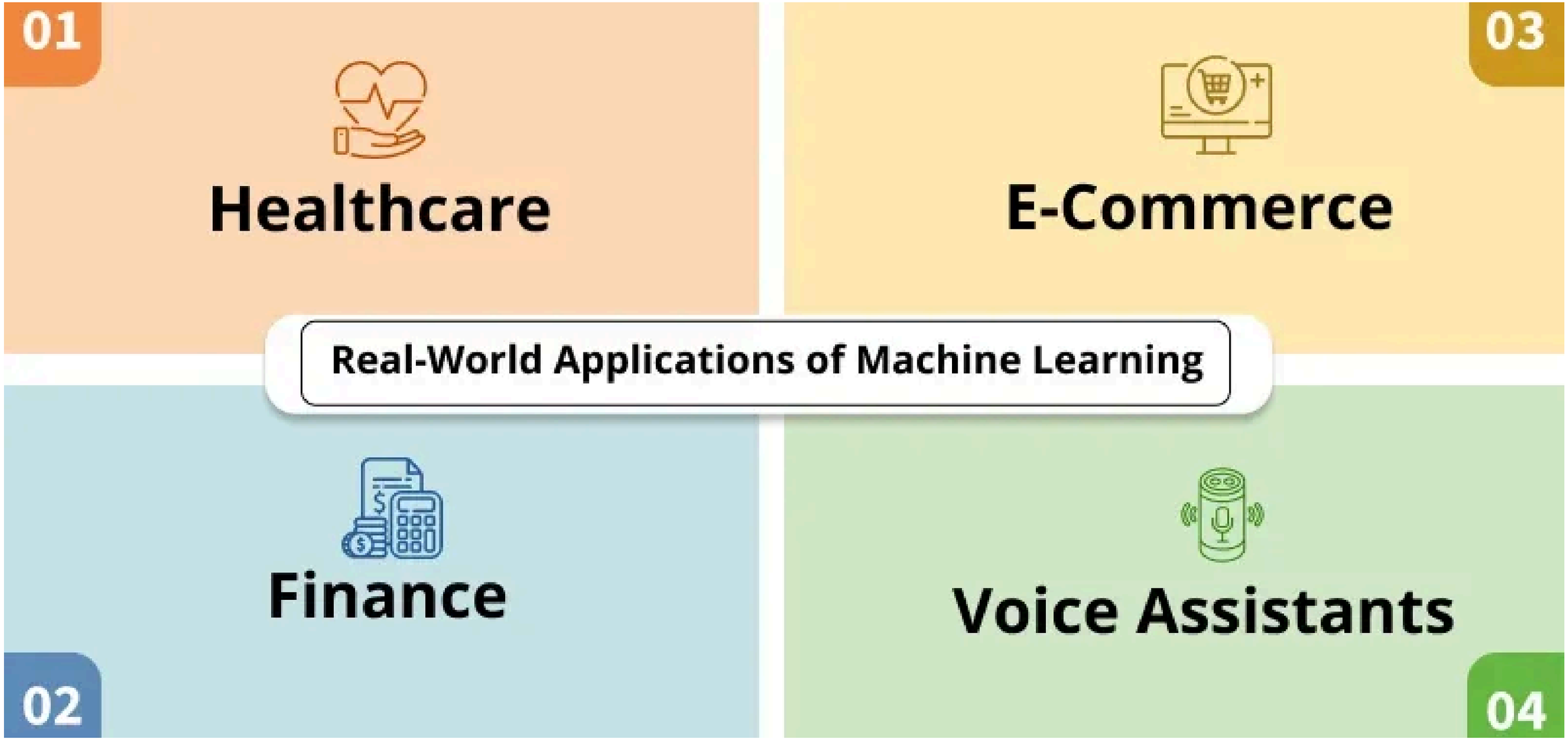


Why do we need Machine Learning?

Traditional programming requires exact instructions and doesn't handle complex tasks like understanding images or language well. It can't efficiently process large amounts of data. Machine Learning solves these problems by learning from examples and making predictions without fixed rules.



Applications of Machine Learning



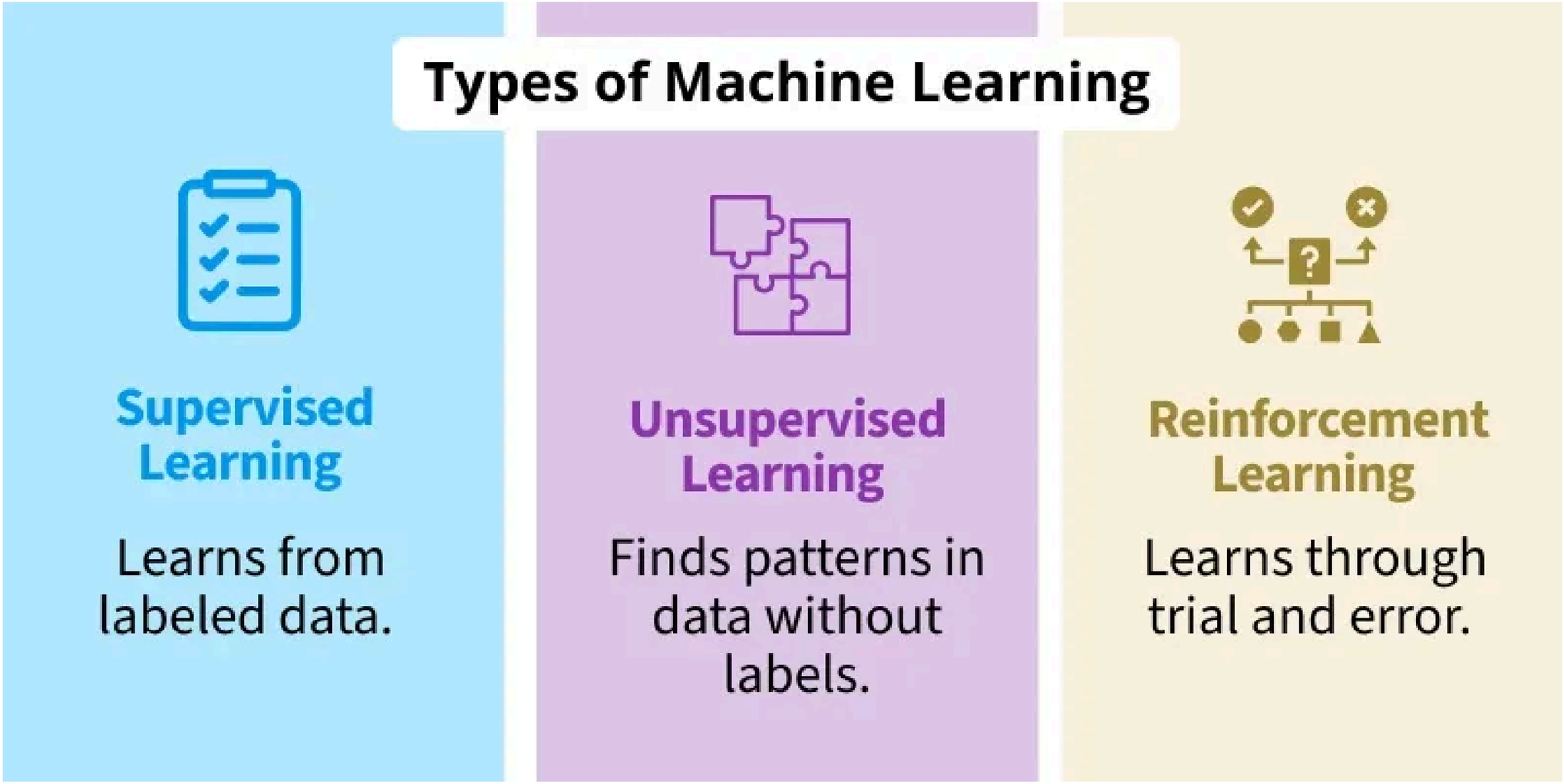
Applications of Machine Learning

Healthcare: It helps doctors to diagnose diseases from medical images like X-rays and MRIs. It also predicts patient outcomes and personalizes treatments which improves healthcare quality.

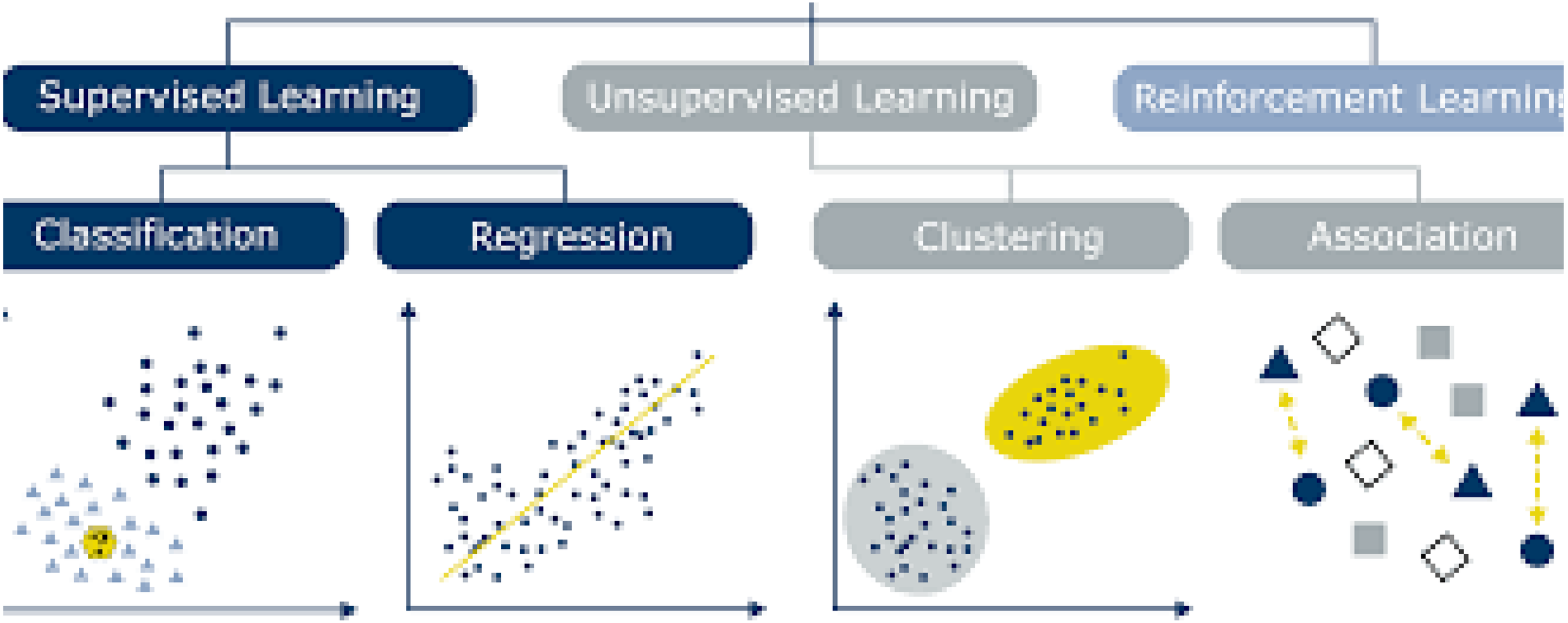
Retail and E-Commerce: It helps in personalized product recommendations and forecasts demand to optimize inventory and also analyzes customer sentiment to improve shopping experiences.

Social Media and Entertainment: Platforms like Netflix and YouTube use ML to recommend content we'll enjoy. It enables image and speech recognition for better user interaction.

Types of Machine Learning



Types of Machine Learning



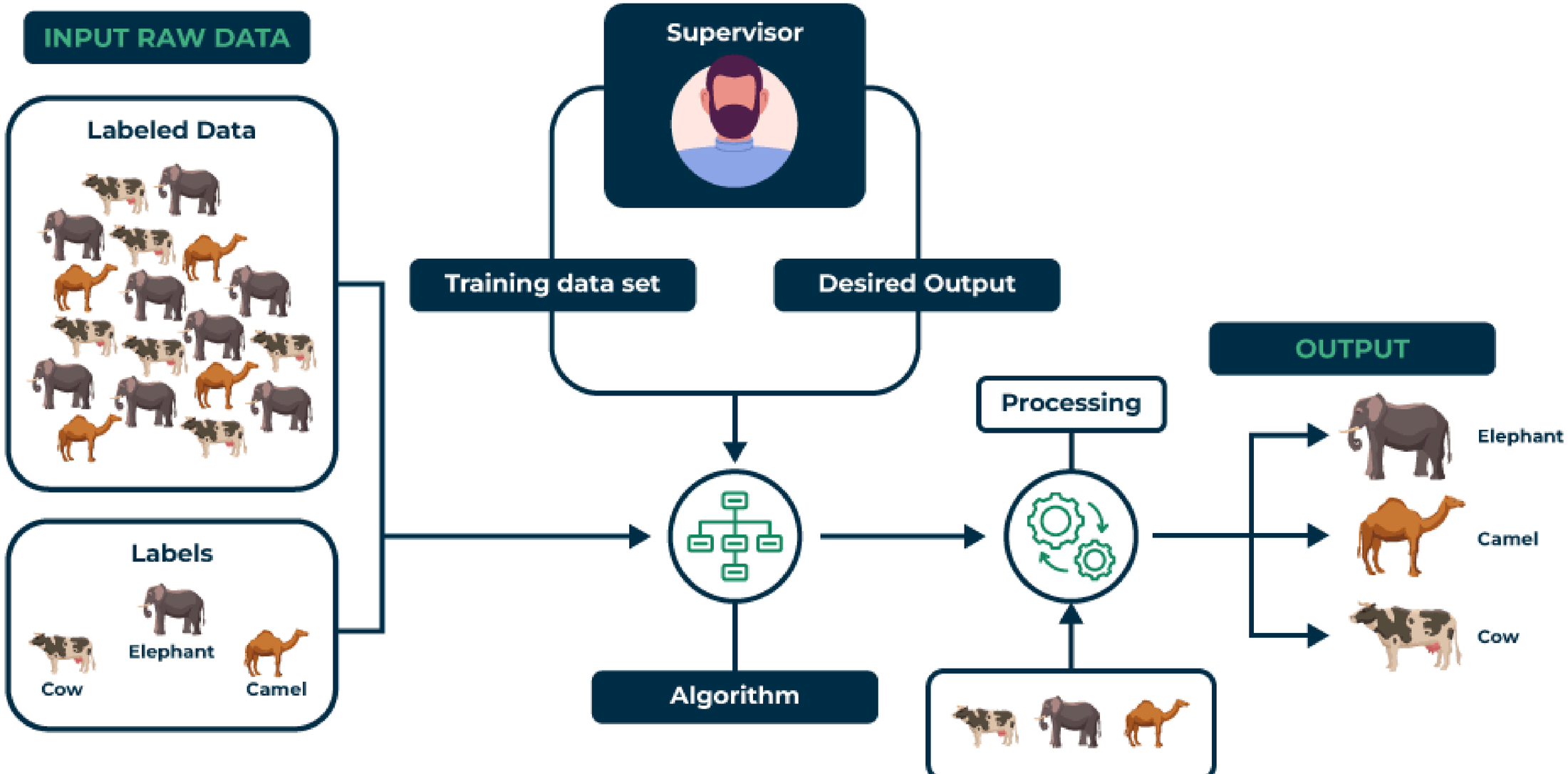
Supervised Machine Learning

Supervised learning is defined as when a model gets trained on a "Labelled Dataset". Labelled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.

There are two main categories of supervised learning

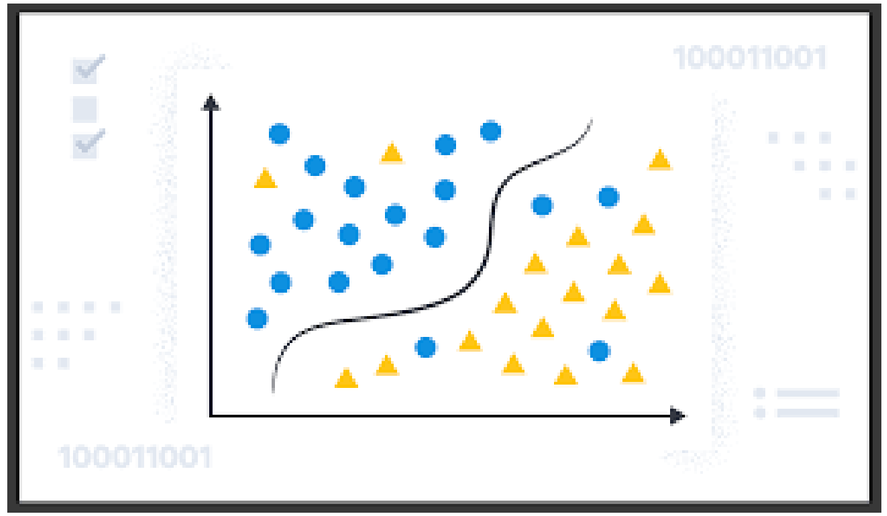
- <u>Classification</u>
- <u>Regression</u>

Supervised Machine Learning

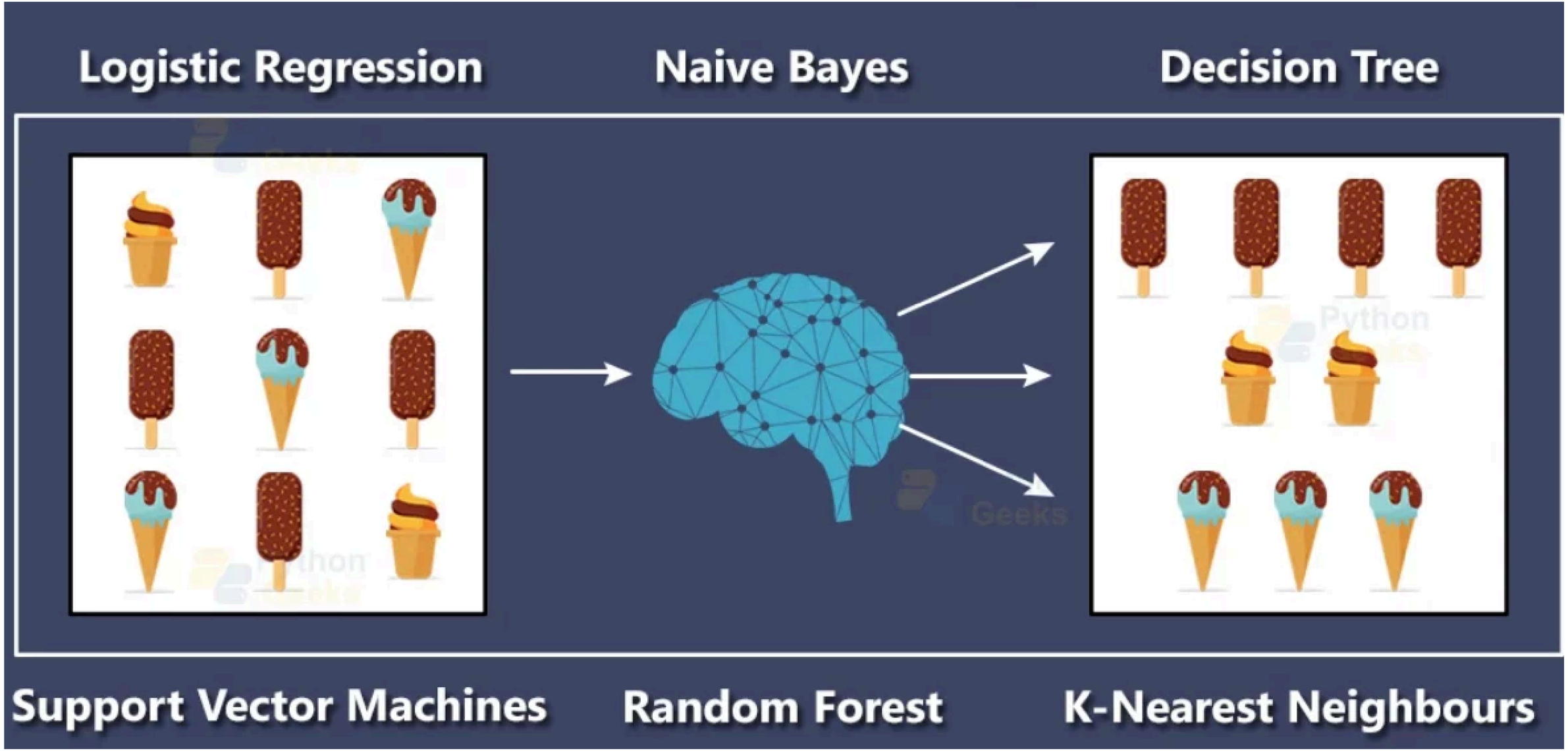


Classification

It deals with predicting categorical target variables, which represent discrete classes or labels. For instance, classifying emails as spam or not spam, or predicting whether a patient has a high risk of heart disease. Classification algorithms learn to map the input features to one of the predefined classes.

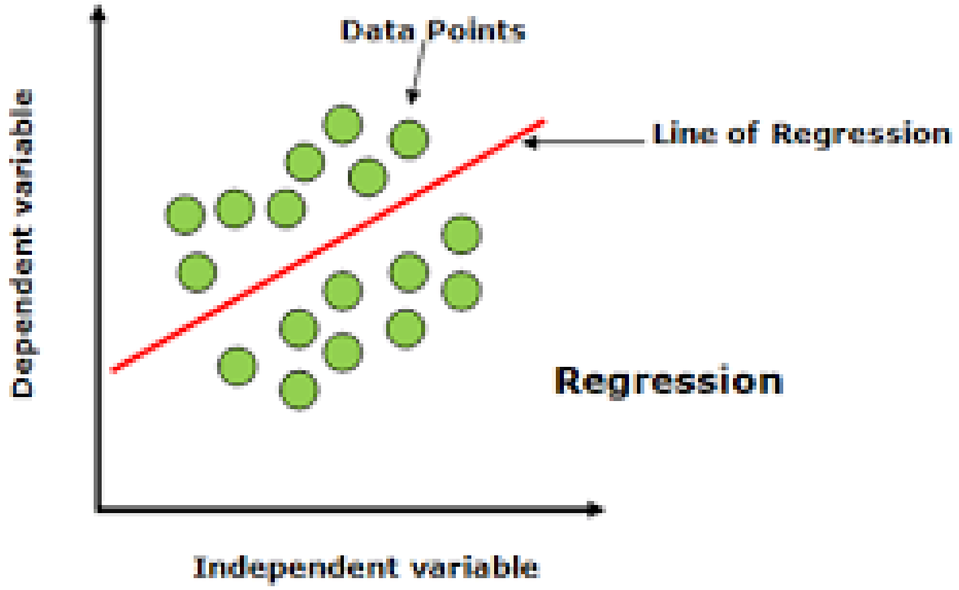


Classification

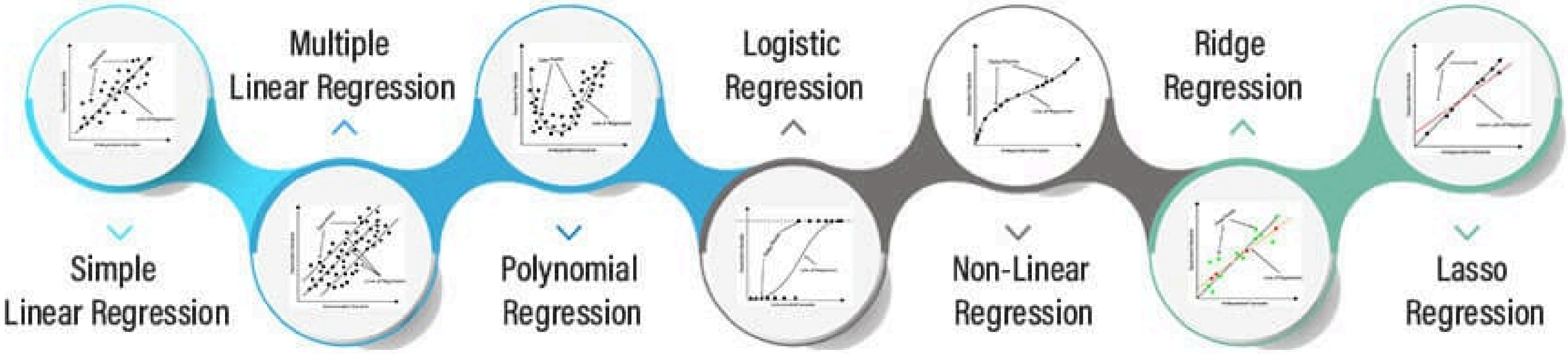


Regression

On the other hand, deals with predicting continuous target variables, which represent numerical values. For example, predicting the price of a house based on its size, location, and amenities, or forecasting the sales of a product. Regression algorithms learn to map the input features to a continuous numerical value.



Regression


Simple
Linear Regression
Multiple
Linear Regression
Polynomial
Regression
Logistic
Regression
Non-Linear
Regression
Ridge
Regression
Lasso
Regression

Difference Between Regression and Classification

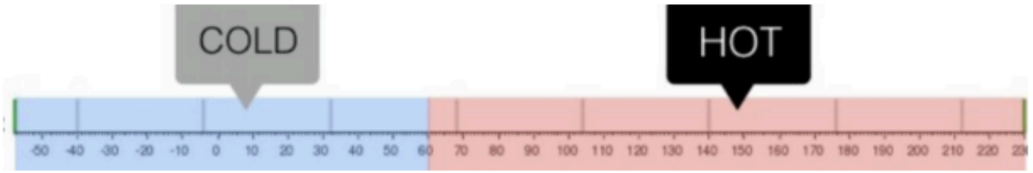
Regression

What will be the temperature tomorrow?

Fahrenheit

Classification

Will it be hot or cold tomorrow?



Fahrenheit

Supervised Learning Algorithms

Regression

- Linear
- Polynomial
- Ridge Regression
- Lasso Regression

Classification

- Logistic Regression
- Linear Discriminant Analysis
- Naïve Bayes

Regression & Classification

- Decision Trees
- K- Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Advantages of Supervised Machine Learning

Supervised Learning models can have high accuracy as they are trained on labelled data.

The process of decision-making in supervised learning models is often interpretable.

It can often be used in pre-trained models which saves time and resources when developing new models from scratch.

Disadvantages of Supervised Machine Learning

- It has limitations in knowing patterns and may struggle with unseen or unexpected patterns that are not present in the training data.
- It can be time-consuming and costly as it relies on labeled data only.
- It may lead to poor generalizations based on new data.

Applications of Supervised Learning

Image classification: Identify objects, faces, and other features in images.

Natural language processing: Extract information from text, such as sentiment, entities, and relationships.

Recommendation systems: Make personalized recommendations to users.

- Predictive analytics: Predict outcomes, such as sales, customer churn, and stock prices.

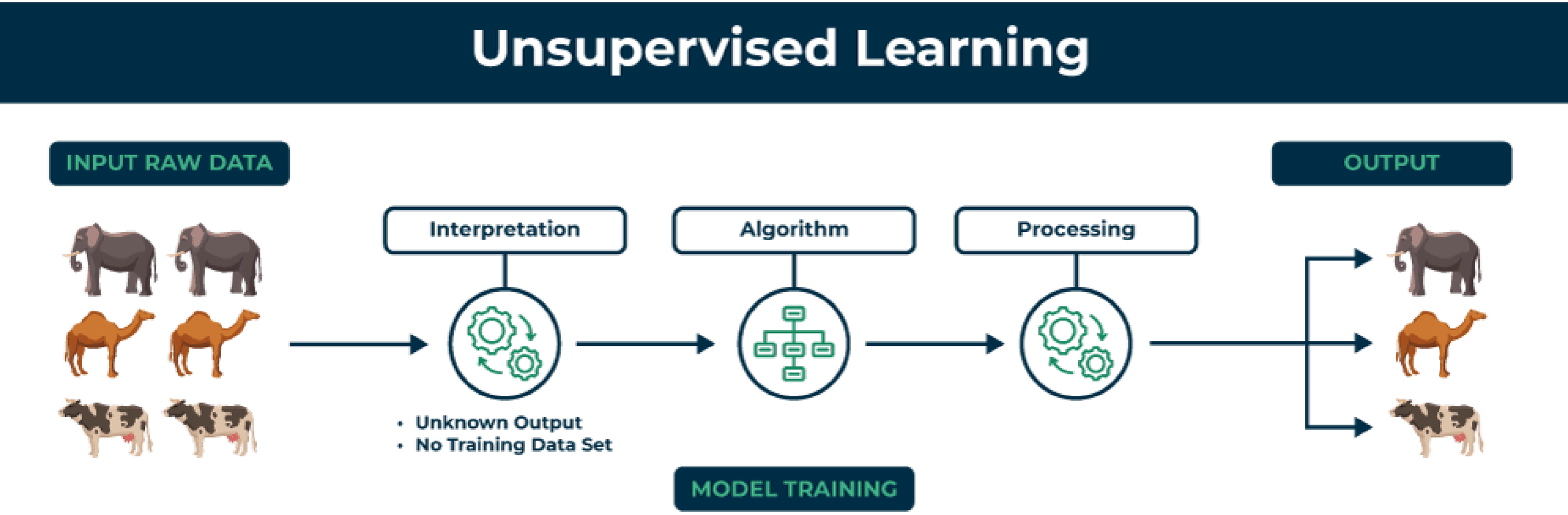
Unsupervised Machine Learning

Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data. Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs. The primary goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.

There are two main categories of unsupervised learning that are mentioned

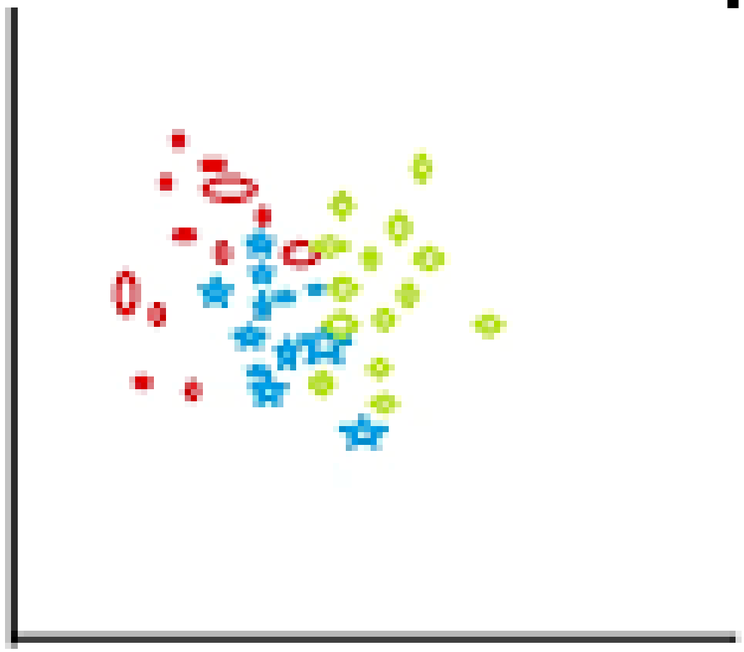
Clustering
Association

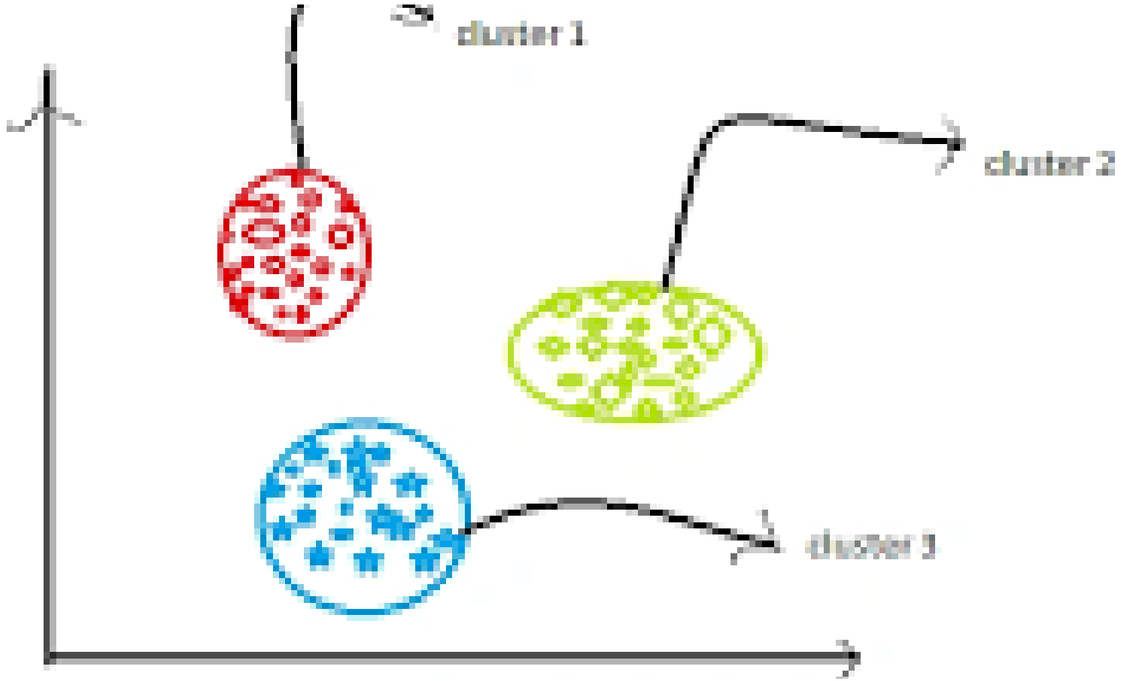
Unsupervised Machine Learning



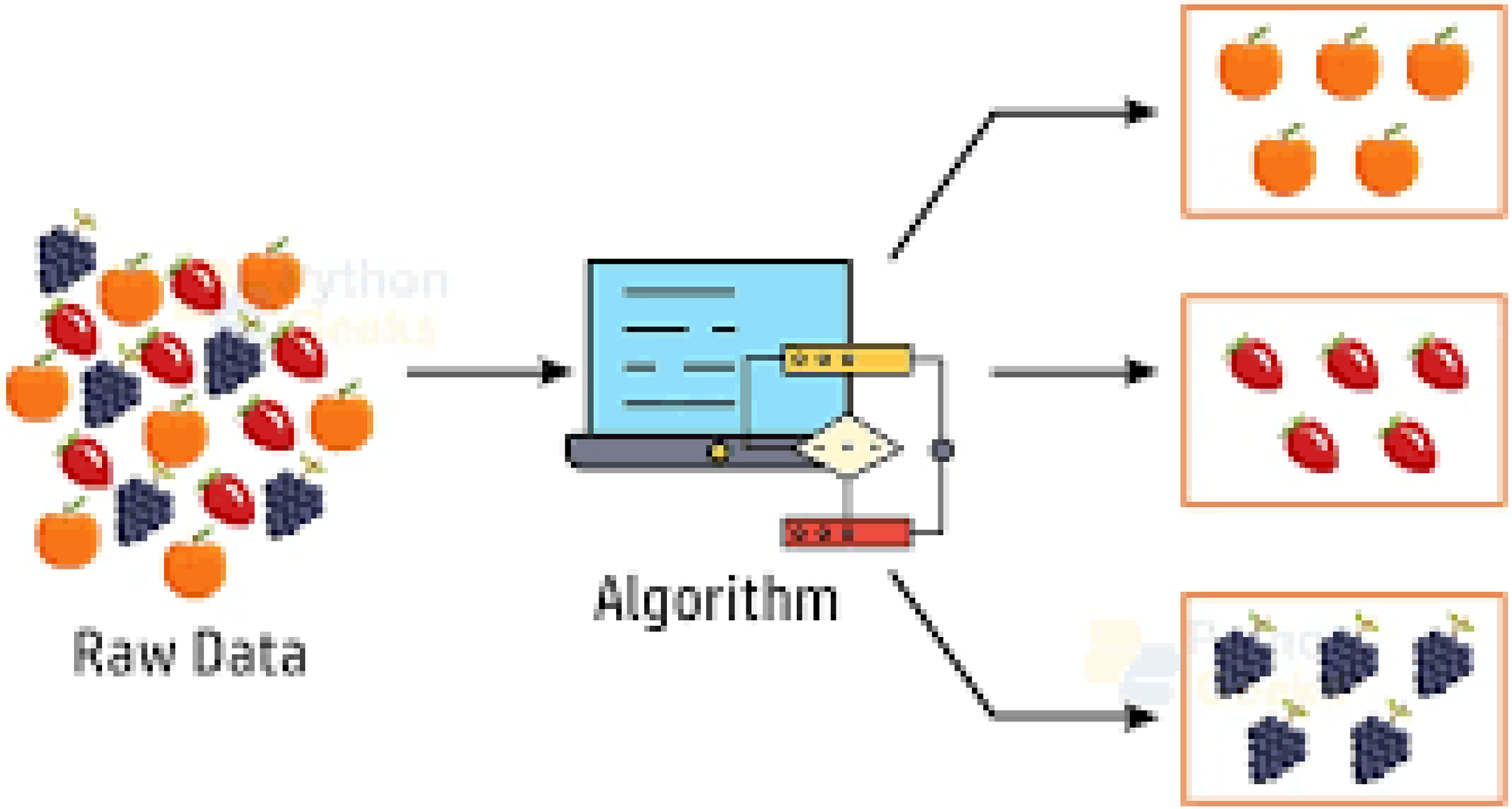
Clustering

Clustering is the process of grouping data points into clusters based on their similarity. This technique is useful for identifying patterns and relationships in data without the need for labeled

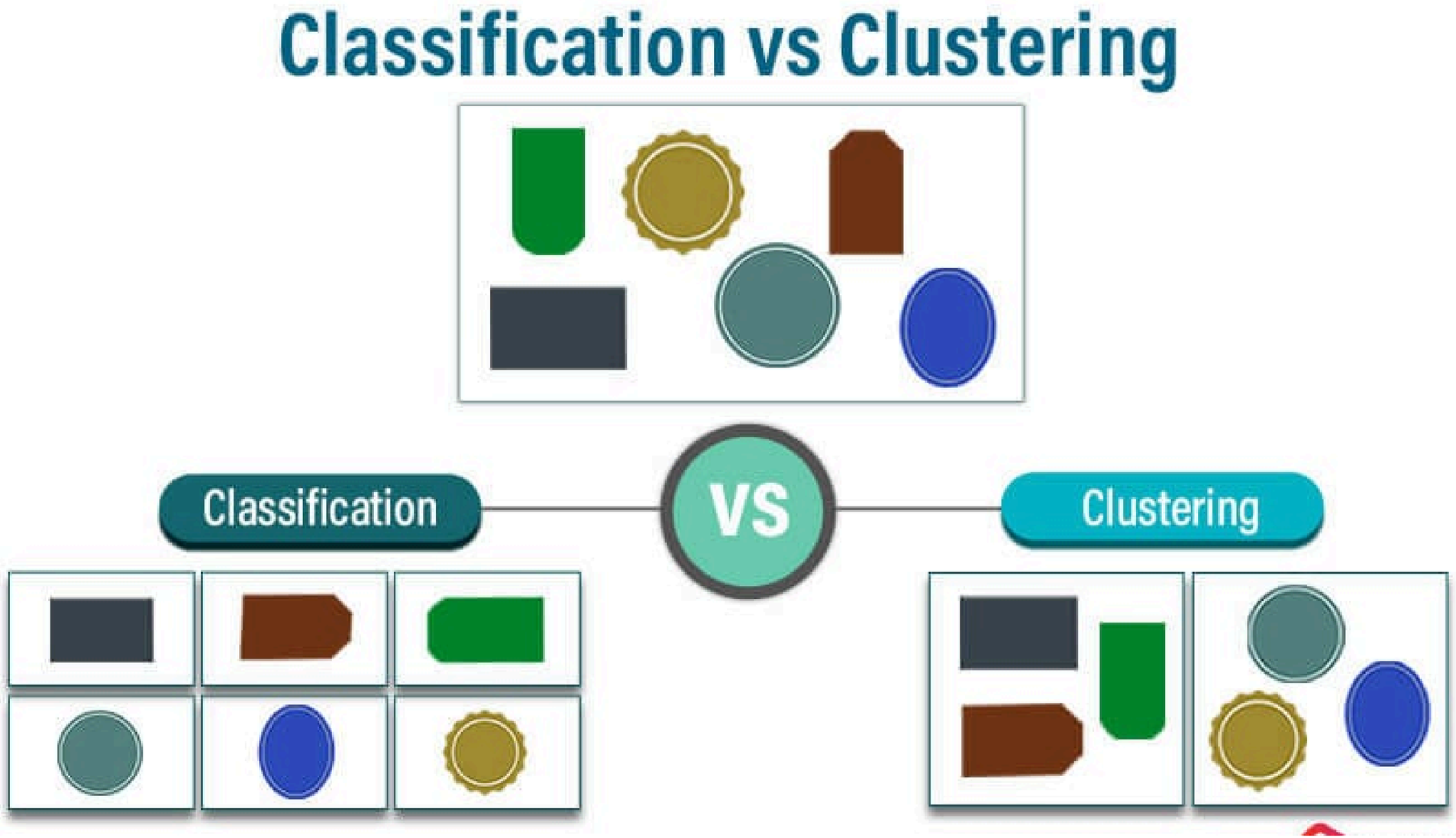




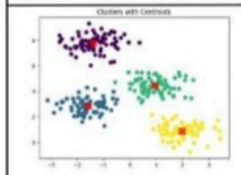
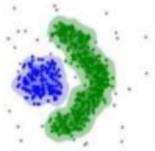
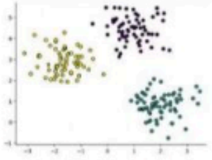
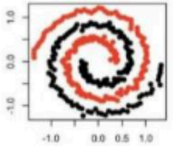
Clustering



Difference Between Classification and Clustering

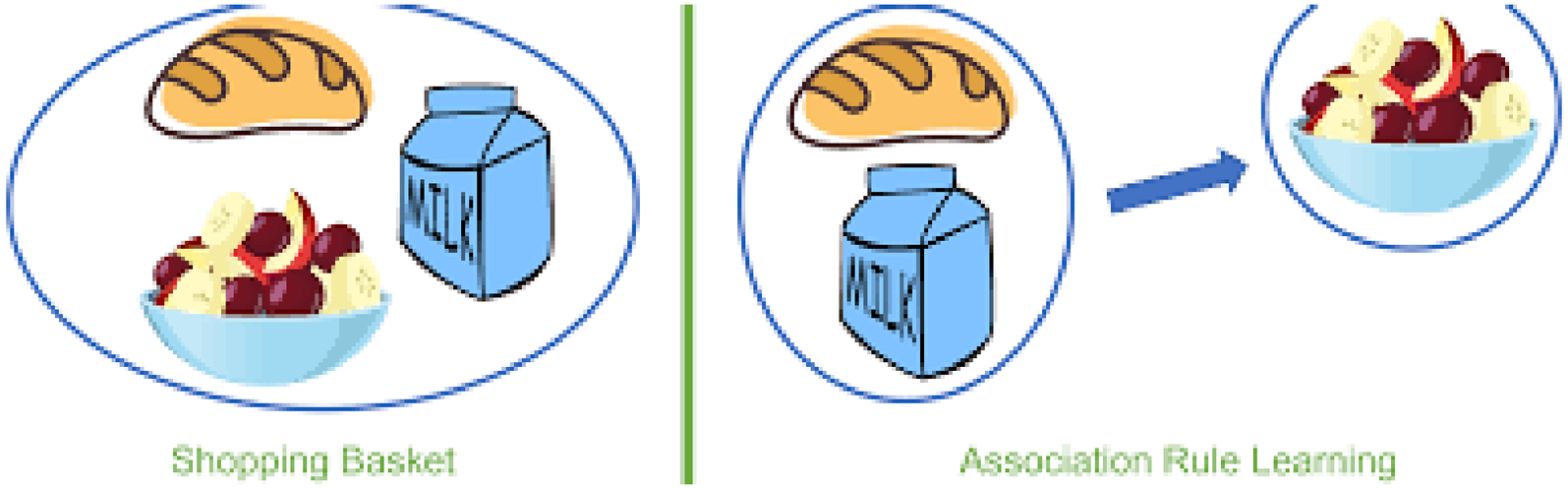


Clustering Algorithms

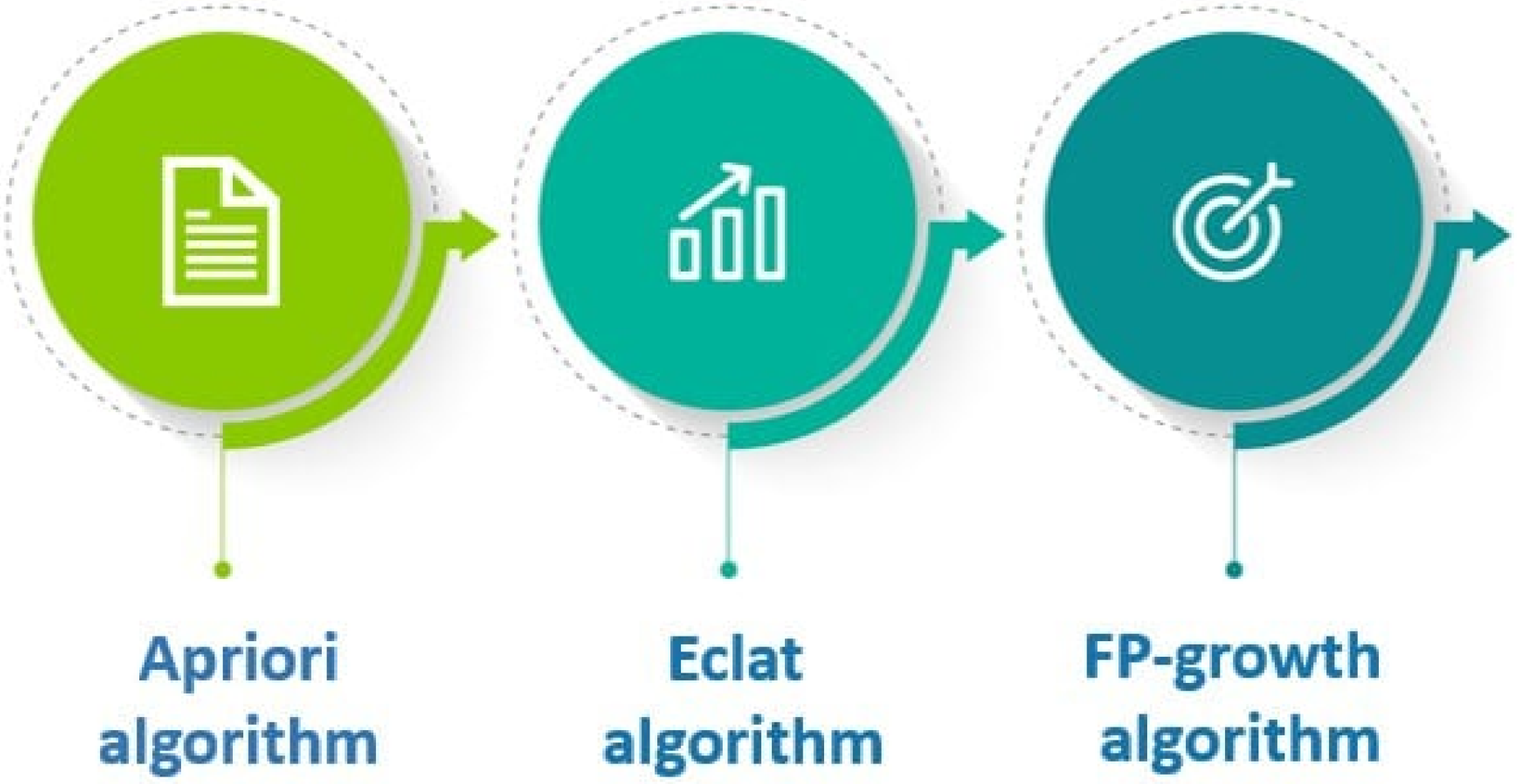
Representation	Algorithm Name	Hyperparameter
	K-Means Clustering	Partitions data into K clusters by minimizing variance within each cluster.
	Hierarchical Clustering	Builds a hierarchy of clusters by iteratively merging or splitting existing groups.
	DBSCAN	Forms clusters based on density; groups densely packed points and marks outliers.
	Mean Shift	Finds clusters by locating and adapting to the centroids of data points.
	Spectral Clustering	Uses eigenvalues of similarity matrix to reduce dimensions before clustering.

Association

Association Rule learning is a technique for discovering relationships between items in a dataset. It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability.



Association



Advantages of Unsupervised Machine Learning

It helps to discover hidden patterns and various relationships between the data.

Used for tasks such as customer segmentation, anomaly detection, and data exploration.

It does not require labeled data and reduces the effort of data labeling.

Disadvantages of Unsupervised Machine Learning

Cluster Interpretability may not be clear and may not have meaningful interpretations.

- Without using labels, it may be difficult to predict the quality of the model's output.

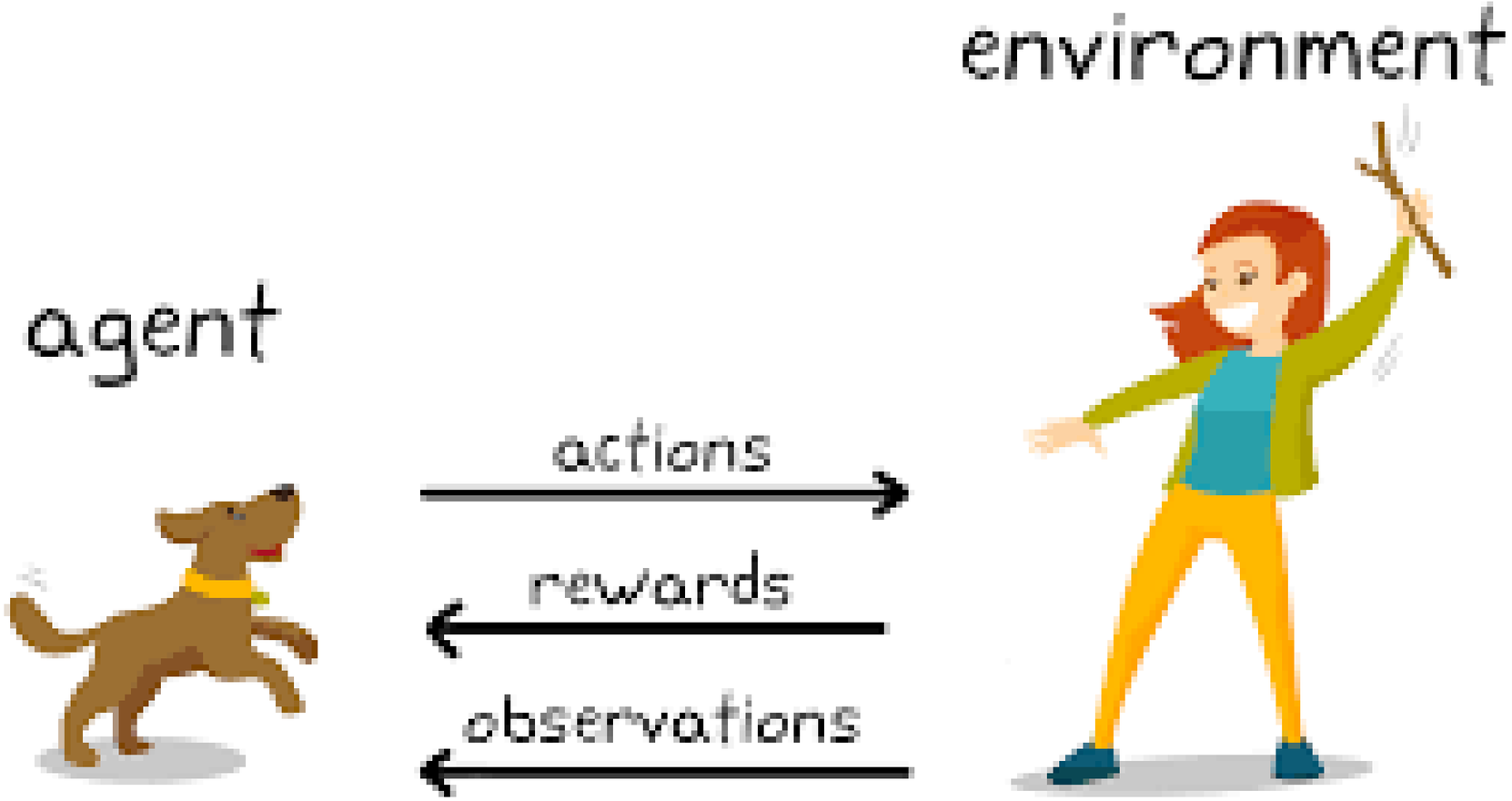
Applications of Unsupervised Learning

- Clustering: Group similar data points into clusters.
- Dimensionality reduction: Reduce the dimensionality of data while preserving its essential information.
- Recommendation systems: Suggest products, movies, or content to users based on their historical behavior or preferences.
- Image segmentation: Segment images into meaningful regions.

Reinforcement Machine Learning

It is a learning method that interacts with the environment by producing actions and discovering errors. Trial, error, and delay are the most relevant characteristics of reinforcement learning. In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern. These algorithms are specific to a particular problem e.g. Google Self Driving car, AlphaGo where a bot competes with humans and even itself to get better and better performers in Go Game. Each time we feed in data, they learn and add the data to their knowledge which is training data. So, the more it learns the better it gets trained and hence experienced.

Reinforcement Machine Learning

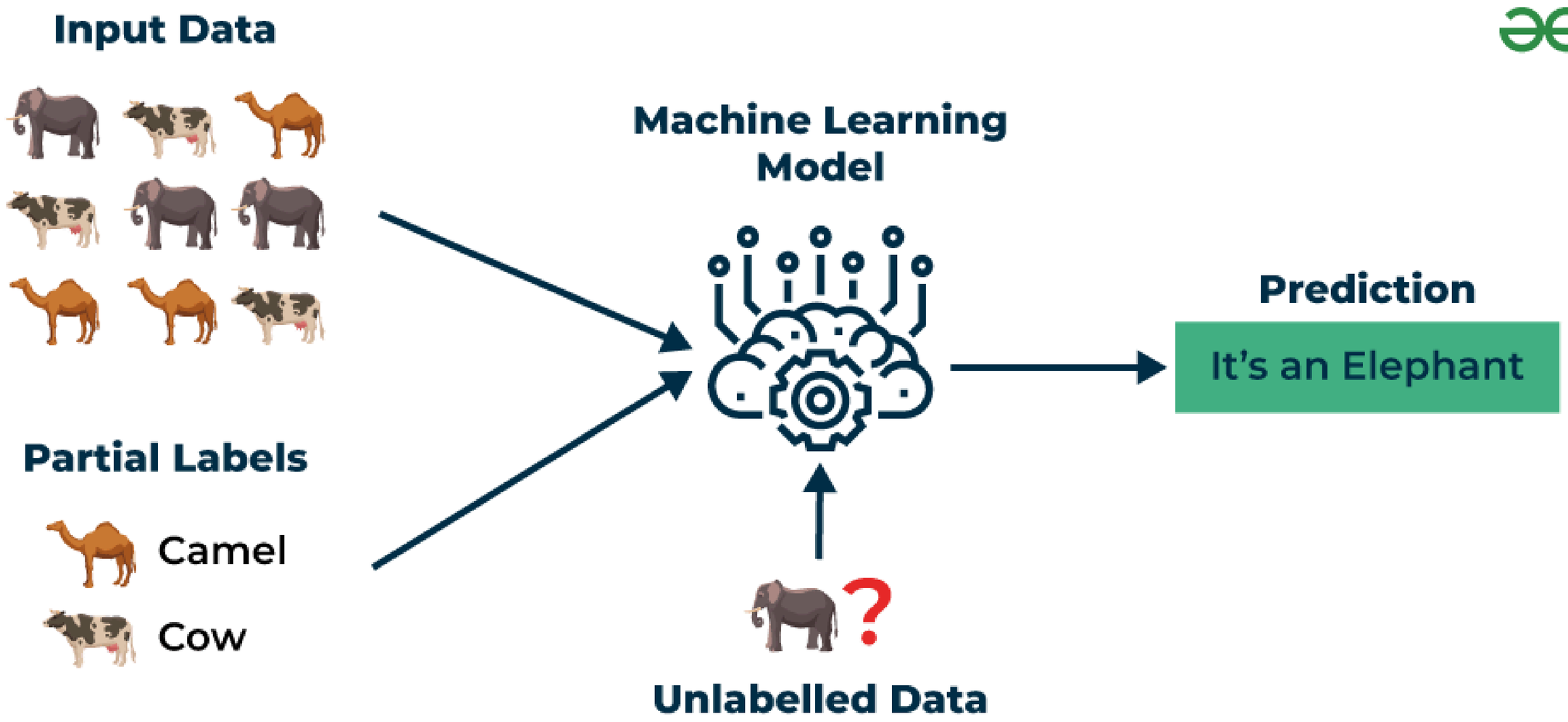


Semi-Supervised Learning: Supervised + Unsupervised Learning

It is a machine learning algorithm that works between the supervised and unsupervised learning so it uses both labelled and unlabelled data. It's particularly useful when obtaining labeled data is costly, time-consuming, or resource-intensive. This approach is useful when the dataset is expensive and time-consuming.

We use these techniques when we are dealing with data that is a little bit labeled and the rest large portion of it is unlabeled. We can use the unsupervised techniques to predict labels and then feed these labels to supervised techniques. This technique is mostly applicable in the case of image data sets where usually all images are not labeled.

Semi-Supervised Learning: Supervised + Unsupervised Learning



Machine Learning Algorithms

Machine learning

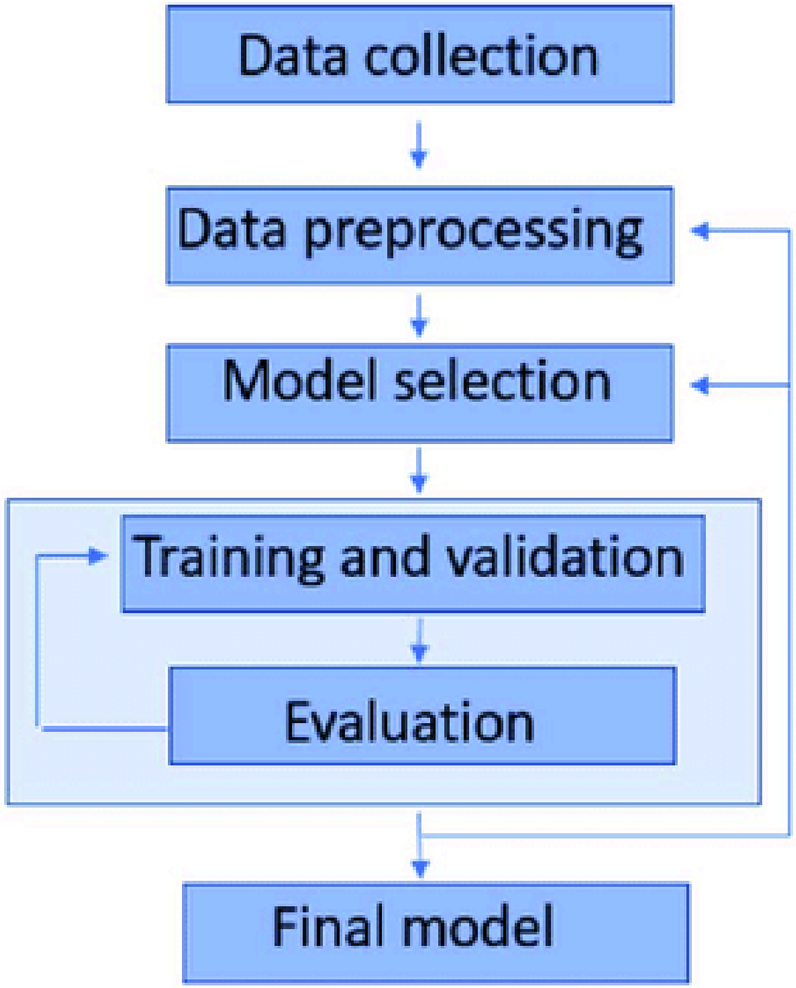
Supervised

- **Regression**
  - Linear regression
  - Generalized linear regression
  - Isotonic regression
  - Survival regression.
- **Classification**
  - Support vector machine
  - Naïve Bayes
  - Logistic regression
  - Decision trees
  - Gradient boosted trees
  - Random forest.

Unsupervised

- **Clustering**
  - K-means
  - Bisecting K-means
  - Streaming K-means
  - Gaussian mixture model.
- **Dimensionality reduction**
  - Principal component analysis
  - Singular value decomposition.

Reinforcement

- Markov decision process
- Q-learning
- Multi-armed bandit.

Machine Learning Pipeline